



AWS Fundamentals and Cloud Practitioner

Learning Summary

Internal Training Documentation

Name

Piyush Kumar

Employee ID

513558

Team / Department

Cloud and Data Center

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1. Introduction

This document summarises the learning outcomes from a structured study of Amazon Web Services (AWS) cloud fundamentals and AWS Cloud Practitioner–style training modules. The learning activity was designed to build a solid foundation in cloud computing before progressing to more advanced topics in cloud networking and AWS architecture.

During this training period, the following areas were studied:

- Understanding the core concepts and benefits of AWS cloud computing.
- Completing AWS Cloud Practitioner–style learning modules covering cloud concepts, core services, security, pricing, and support.
- Learning the basic structure of an Amazon Virtual Private Cloud (VPC).
- Understanding subnets, routing, Security Groups (SGs), and Network Access Control Lists (NACLs).
- Preparing technical notes and a basic VPC architecture diagram to consolidate understanding.

The purpose of this learning was to build foundational cloud knowledge that will support future study of advanced cloud networking, multi-account architectures, and AWS solution design.

2. AWS Cloud Fundamentals

2.1 Cloud Computing

Cloud computing refers to the delivery of computing resources over the internet on a pay-as-you-go basis. Rather than purchasing and maintaining physical hardware, organisations can access infrastructure and services hosted and managed by a cloud provider. Cloud computing typically includes the following categories of resources:

- Compute – virtual servers and processing power for running applications.
- Storage – scalable data storage for files, objects, and archives.
- Databases – managed relational and non-relational database services.
- Networking – virtual networks, load balancers, and connectivity services.
- Security – identity management, encryption, and access controls.
- Monitoring – tools for tracking performance, usage, and system health.
- Application services – serverless functions, queuing, notifications, and APIs.

The key benefits of cloud computing include:

- Scalability – resources can be scaled up or down to match demand.
- Flexibility – a wide range of services can be selected to suit different workloads.
- High availability – services are designed to remain accessible even during component failures.
- Pay-as-you-go pricing – organisations pay only for the resources they consume.

- Faster resource provisioning – new infrastructure can be deployed in minutes rather than weeks.
- Reduced dependency on physical infrastructure – eliminates the need to purchase and manage on-premises hardware.

2.2 Amazon Web Services

Amazon Web Services, commonly known as AWS, is one of the world's leading cloud platforms. AWS provides a broad portfolio of infrastructure and managed services across multiple geographic locations, referred to as Regions and Availability Zones (AZs). These services enable organisations of all sizes to build, deploy, and scale applications with speed and reliability. AWS supports use cases ranging from web hosting and data analytics to machine learning and enterprise application migration.

3. AWS Cloud Practitioner–Style Modules

The AWS Cloud Practitioner–style curriculum is an introductory training track that covers four major knowledge domains. Each domain provides foundational understanding of a key aspect of the AWS platform.

3.1 Cloud Concepts

This module introduced the core ideas that define cloud computing and the AWS approach to delivering services. The main concepts studied include:

- Benefits of cloud computing – the advantages of using cloud services over on-premises infrastructure.
- Scalability and elasticity – the ability to increase capacity (scale out) or reduce it (scale in) automatically or on demand.
- High availability – designing systems that continue to operate even when individual components fail.
- Fault tolerance – the ability of a system to continue functioning correctly in the event of a partial failure.
- Agility – the speed at which organisations can provision resources and experiment with new ideas.
- Global reach – AWS operates Regions and AZs around the world, enabling low-latency access for global users.
- Shared responsibility model – AWS is responsible for the security of the cloud infrastructure; customers are responsible for securing the resources and data they place in the cloud.

3.2 AWS Core Services

This module provided an introductory overview of the most commonly used AWS services. The table below summarises the key services covered during training:

AWS Service	Purpose
Amazon EC2 (Elastic Compute Cloud)	Provides resizable virtual servers in the cloud for running applications.
Amazon S3 (Simple Storage Service)	Scalable object storage for files, backups, and static content.
Amazon RDS (Relational Database Service)	Managed relational database service supporting engines such as MySQL and PostgreSQL.
AWS Lambda	Serverless compute service that runs code in response to events without provisioning servers.
Amazon VPC (Virtual Private Cloud)	Enables you to define an isolated network within AWS.
AWS IAM (Identity and Access Management)	Controls who can access AWS resources and what actions they may perform.
Amazon CloudWatch	Monitoring and observability service for AWS resources and applications.

3.3 Security and Compliance

Cloud security is a shared responsibility between AWS and the customer. This module covered the following important security principles:

- Identity and access management – ensuring that only authorised users and services can access AWS resources.
- Least-privilege access – granting users and applications only the minimum permissions required to perform their tasks.
- Multi-factor authentication (MFA) – requiring a second form of verification in addition to a password for critical accounts.
- Data encryption – protecting data at rest and in transit using industry-standard encryption.
- Logging and monitoring – recording and reviewing access and activity logs to detect unusual behaviour.
- AWS shared responsibility model – understanding which security controls are managed by AWS and which are managed by the customer.
- Compliance awareness – AWS provides services and documentation to assist with regulatory and industry compliance requirements.

Key Principle: Shared Responsibility Model

AWS manages security **OF** the cloud (physical infrastructure, hypervisor, network hardware).
Customers manage security **IN** the cloud (operating systems, applications, data, access controls).

3.4 AWS Pricing and Support

Understanding AWS pricing is important for managing cloud costs effectively. The following pricing and support concepts were covered:

- Pay-as-you-go pricing – costs are based on actual usage; there are no upfront commitments by default.
- On-demand resources – resources that are provisioned and billed by the hour or second with no long-term commitment.
- Savings Plans – flexible pricing models that offer lower rates in exchange for a commitment to a consistent usage level over one or three years.
- Reserved capacity – allows organisations to reserve specific resource capacity at a discounted rate compared to on-demand pricing.
- AWS Free Tier – provides limited access to many AWS services at no charge, enabling new users to learn and experiment.
- AWS Cost Explorer – a tool for visualising and analysing AWS usage and costs over time.
- AWS Budgets – allows organisations to set spending thresholds and receive alerts when costs approach or exceed defined limits.
- AWS Support Plans – AWS offers several tiers of support, ranging from basic documentation access to dedicated technical account management for enterprise customers.

Note on Pricing

Exact pricing figures are not included in this document because AWS pricing is subject to change.

Always refer to the official AWS Pricing page or AWS Cost Calculator for current rates.

4. Amazon Virtual Private Cloud

An Amazon Virtual Private Cloud, or Amazon VPC, is a logically isolated network within the AWS cloud. A VPC allows an organisation to define a private network boundary in which AWS resources such as servers, databases, and load balancers can be deployed securely. The organisation has full control over the VPC's network configuration.

The main purposes and capabilities of a VPC include:

- Defining a private network boundary within AWS.
- Selecting a custom IP address range using a Classless Inter-Domain Routing (CIDR) block.
- Creating subnets within the VPC to segment the network.
- Configuring route tables to control traffic routing between subnets and to external networks.
- Controlling inbound and outbound traffic using Security Groups and NACLs.
- Connecting resources to the internet, to other VPCs, or to on-premises networks.

A Classless Inter-Domain Routing block, commonly called a CIDR block, defines the IP address range available within a VPC or subnet. For example, a CIDR block of 10.0.0.0/16 provides 65,536 IP addresses, while a subnet CIDR of 10.0.1.0/24 provides 256 addresses within a smaller segment of that range.

** Learning Example: CIDR Ranges**

VPC CIDR block: 10.0.0.0/16 (provides 65,536 addresses)

Public subnet CIDR: 10.0.1.0/24 (256 addresses in the public subnet)

Private subnet CIDR: 10.0.2.0/24 (256 addresses in the private subnet)

Note: The above example is for learning purposes only and does not represent a specific production environment.

5. Subnets

A subnet is a smaller IP address range created within a VPC. Subnets are used to divide the VPC's address space into separate segments, each serving a specific purpose. Subnets are associated with Availability Zones and are classified as either public or private based on their routing configuration.

5.1 Public Subnet

A public subnet is a subnet that has a route to an Internet Gateway (IGW), which allows resources within it to communicate with the internet. Resources typically placed in a public subnet include:

- Public-facing web servers that must be accessible to external users.
- Load balancers that distribute incoming internet traffic.
- Bastion hosts (also called jump servers) used for secure administrative access.
- Network Address Translation (NAT) Gateways that enable private resources to reach the internet.

** Important Clarification**

Placing a resource in a public subnet does not automatically make it publicly accessible.

Public accessibility also depends on whether the resource has a public IP address, whether the route table includes a route to the Internet Gateway, and whether the Security Group and NACL rules permit the relevant traffic.

5.2 Private Subnet

A private subnet does not have a direct route from the internet. Resources in a private subnet cannot be reached directly from outside the VPC, which makes private subnets suitable for sensitive or internal workloads. Resources commonly placed in a private subnet include:

- Application servers that process business logic.
- Internal microservices and APIs.
- Databases and data stores.
- Backend systems that do not require direct internet access.

Private resources that require outbound internet access (for example, to download software updates) can route their traffic through a NAT Gateway located in a public subnet. The NAT

Gateway forwards the request to the internet and returns the response, without allowing inbound internet connections to be initiated to the private resources.

5.3 Subnet and Availability Zone Relationship

Each subnet in a VPC belongs to exactly one Availability Zone. An Availability Zone (AZ) is an isolated data centre location within an AWS Region. By deploying resources across multiple subnets in different AZs, architects can design systems that remain available even if one AZ experiences an outage. This multi-AZ approach is a common pattern for achieving high availability and resiliency in AWS environments.

6. Routing in AWS

Routing determines where network traffic is directed within and beyond a VPC. AWS uses route tables, an Internet Gateway, and a NAT Gateway to manage the flow of traffic between subnets and external networks.

6.1 Route Table

A route table is a set of rules, known as routes, that determine where network packets are sent. Each route specifies:

- Destination – the IP address range that the rule applies to (for example, 0.0.0.0/0 for all internet traffic).
- Target – the resource to which matching traffic should be forwarded (for example, an Internet Gateway or NAT Gateway).

Common routing examples include:

- Local VPC traffic is automatically routed using the local route, which covers all addresses within the VPC CIDR block.
- Internet-bound traffic from a public subnet is routed to the Internet Gateway.
- Internet-bound traffic from a private subnet is routed through a NAT Gateway in the public subnet.

6.2 Main Route Table

Every VPC is automatically created with a main route table. By default, all subnets that have not been explicitly associated with a custom route table will use the main route table. Separate custom route tables can be created and associated with specific subnets to apply different routing rules, such as providing internet access to public subnets while keeping private subnets isolated.

6.3 Internet Gateway

An Internet Gateway (IGW) is a horizontally scaled, redundant, and highly available VPC component that enables communication between a VPC and the internet. For internet access to work, the following must be configured correctly:

- The IGW must be attached to the VPC.
- The subnet's route table must contain a route directing internet traffic (0.0.0.0/0) to the IGW.
- The resource must have a public or Elastic IP address.

- Security Group and NACL rules must permit the relevant inbound and outbound traffic.

6.4 NAT Gateway

A NAT Gateway (Network Address Translation Gateway) enables resources in a private subnet to initiate outbound connections to the internet while preventing unsolicited inbound connections from the internet. The NAT Gateway is placed in a public subnet and translates the private IP addresses of outbound requests to a public IP address.

The table below summarises the routing components covered in this section:

Component	Main Purpose	Common Placement
Internet Gateway	Connects the VPC to the internet	Attached to the VPC
NAT Gateway	Provides outbound internet access for private resources	Public subnet
Route Table	Controls traffic destinations	Associated with subnets

7. Security Groups

A Security Group (SG) is a virtual firewall that controls inbound and outbound traffic for AWS resources such as Amazon EC2 (Elastic Compute Cloud) instances. Each Security Group is associated with specific resources at the network-interface level.

The key characteristics of Security Groups are:

- They operate at the resource or network-interface level, not at the subnet level.
- They support separate inbound and outbound rules.
- They are stateful, meaning that return traffic is automatically permitted for allowed inbound connections.
- They allow traffic only when a matching allow rule exists; there are no explicit deny rules.

Stateful behaviour explained: if an inbound rule allows HTTPS traffic from the internet on port 443, the response traffic returning to the client is automatically permitted, even if no outbound rule has been defined for that specific response.

Practical examples of Security Group rules:

- Allow HTTPS traffic on port 443 from any IP address (0.0.0.0/0) for public-facing web servers.
- Allow SSH (Secure Shell) access on port 22 only from a specific approved administrative IP address.
- Allow an application server to connect to a database on port 3306 (MySQL) from within the VPC.

⚠ Security Best Practice

Broad access rules using 0.0.0.0/0 should be used only when genuinely required and must be carefully reviewed.

Where possible, restrict rules to specific IP addresses, CIDR ranges, or source Security Groups.

8. Network Access Control Lists

A Network Access Control List (NACL) is an optional, stateless security control that operates at the subnet level within a VPC. While Security Groups protect individual resources, NACLs provide an additional layer of protection by filtering traffic entering and leaving entire subnets.

The key characteristics of NACLs are:

- They operate at the subnet level, applying to all traffic crossing the subnet boundary.
- They support both inbound and outbound rules.
- They are stateless, meaning that return traffic must be explicitly permitted by a separate outbound rule.
- They support both allow and deny rules, providing more granular control than Security Groups.
- Rules are processed in ascending numerical order; the first matching rule is applied.
- Return traffic must be explicitly permitted by an outbound rule.

Stateless behaviour explained: if an inbound NACL rule allows HTTPS traffic on port 443, a corresponding outbound rule must also be configured to allow the response traffic to return to the client. Without the outbound rule, the response will be blocked at the subnet boundary.

9. Security Group and NACL Comparison

The table below compares the key characteristics of Security Groups and NACLs to help clarify how each control operates and when each is typically used:

Feature	Security Group (SG)	Network ACL (NACL)
Applied At	Resource or network-interface level	Subnet level
Behaviour	Stateful	Stateless
Rule Types	Allow rules only	Allow and deny rules
Rule Processing	All rules are evaluated	Rules processed in numerical order
Return Traffic	Automatically permitted for allowed connections	Must be explicitly permitted
Typical Use	Primary resource-level protection	Additional subnet-level protection

Security Groups and NACLs are complementary controls and are often used together to provide layered network security. Security Groups serve as the primary defence at the resource level, while NACLs provide an additional line of defence at the subnet boundary. Using both controls together follows the principle of defence in depth.

10. Basic VPC Architecture Diagram

The diagram below illustrates a basic AWS VPC architecture containing one public subnet and one private subnet. It shows the key networking and security components covered in this learning summary, along with the primary traffic flow paths.

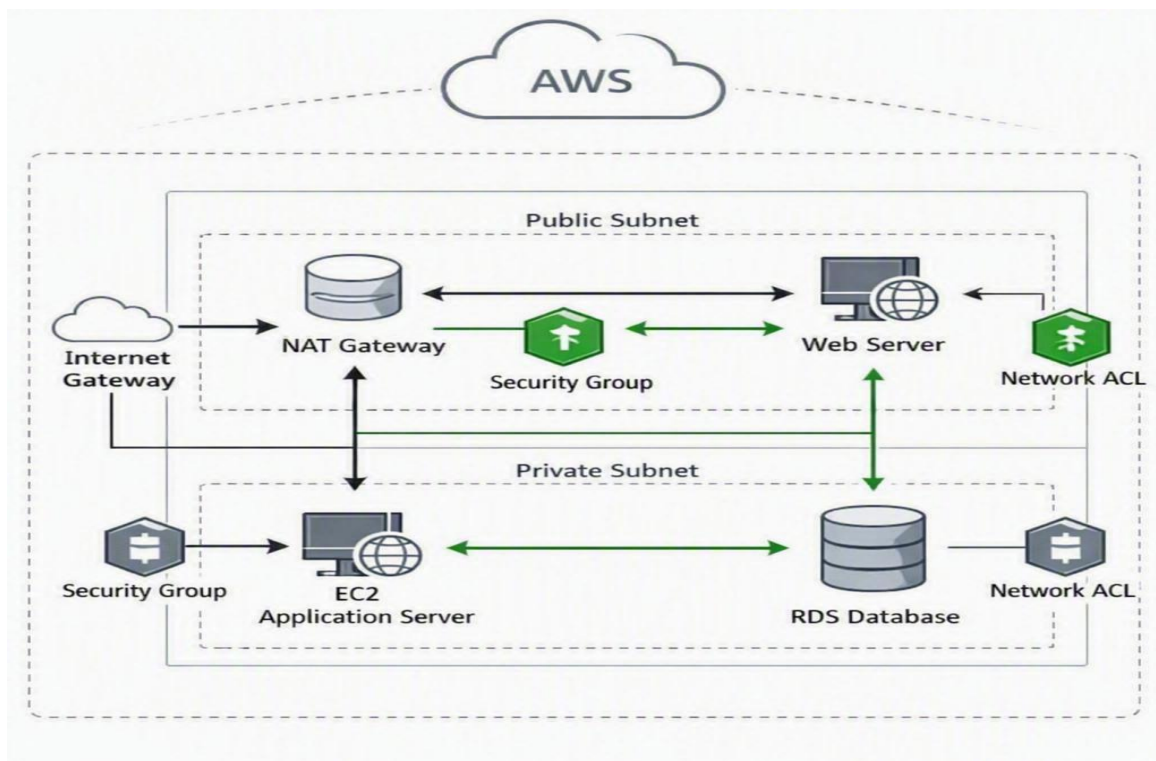


Figure 1: Basic AWS VPC Architecture with Public and Private Subnets

Traffic flow summary:

- Internet traffic enters the VPC through the Internet Gateway.
- The Internet Gateway forwards permitted traffic to the public subnet.
- The public web server communicates with the private application server across the subnet boundary.
- The private application server communicates with the database within the private subnet.
- Private resources use the NAT Gateway for permitted outbound internet access, such as software updates.

11. Example Traffic Flow

The following step-by-step example describes how a user's HTTPS request travels through a basic VPC architecture to reach an application and return a response. This is a simplified learning example and does not represent a complete production architecture.

1. A user sends an HTTPS request from a web browser over the internet.
2. The request arrives at the Internet Gateway attached to the VPC.
3. The public subnet's route table directs the traffic from the IGW toward the web server.
4. The Security Group on the web server allows inbound HTTPS traffic on port 443.
5. The web server processes the request or forwards it to the private application server on an internal port.
6. The application server receives the request and queries the database using an approved internal database port.
7. Security Groups and NACLs control traffic at different layers: Security Groups protect each resource, while NACLs filter traffic at the subnet boundary.
8. The response is returned to the user's browser via the same path in reverse.

12. Hands-On Learning Notes

The following practical activities were completed or reviewed as part of this training. Where applicable, placeholders have been included for personal notes and observations.

- Reviewed the AWS Management Console – explored the navigation structure and identified key service categories.
- Identified AWS Regions and Availability Zones – reviewed the global infrastructure map and noted the purpose of Regions and AZs.
- Understood VPC CIDR ranges – practised reading and interpreting CIDR notation for VPCs and subnets.
- Reviewed public and private subnet design – examined the differences in routing configuration and typical resource placement.
- Reviewed route tables – examined the local route, internet route, and NAT Gateway route entries.
- Understood Internet Gateway and NAT Gateway usage – compared the purpose and placement of each component.
- Compared Security Groups and NACLs – studied the differences in scope, statefulness, and rule types.
- Drew a basic VPC architecture diagram – created a labelled diagram showing the components and traffic flow described in this document.
- Reviewed AWS Cloud Practitioner introductory concepts – studied cloud concepts, core services, security principles, and pricing models.

13. Key Takeaways

The following learning outcomes were achieved through this training activity:

- Gained a clear understanding of the purpose of AWS and the core benefits of cloud computing.

- Learned the four major knowledge areas covered by AWS Cloud Practitioner–style training.
- Understood the purpose and structure of an Amazon VPC, including CIDR blocks and network boundaries.
- Understood the difference between public and private subnets and their typical use cases.
- Learned how route tables direct network traffic within a VPC and to external networks.
- Understood the role of Internet Gateways and NAT Gateways in enabling internet connectivity.
- Understood how Security Groups protect individual AWS resources as stateful virtual firewalls.
- Understood how NACLs provide stateless, subnet-level network security controls.
- Recognised the difference between stateful (Security Groups) and stateless (NACLs) security controls.
- Understood the basic flow of traffic through a VPC from the internet to an application and back.
- VPC Flow Logs – capturing and reviewing network traffic logs for security and troubleshooting.
- AWS Network Firewall – a managed firewall service for fine-grained network traffic filtering.
- AWS Identity and Access Management (IAM) – advanced policies, roles, and permission boundaries.
- Amazon CloudWatch – advanced monitoring, dashboards, and alerting for AWS resources.
- AWS cost optimisation strategies – Savings Plans, Reserved Instances, and cost allocation tags.
- Hands-on VPC creation using the AWS Management Console.
- Infrastructure as Code (IaC) – automating AWS resource provisioning using AWS CloudFormation or Terraform.